Subobject Classifier (polish)

intro

- somehow you are interesting in topos and so subobject classifier
- definition of subobject classifier in wiki

define subobject

- factor through
- equivalence relation

example

- greatest subobject $\mathrm{id}_A : A \hookrightarrow A$
- diagonal embedding $\Delta_A \coloneqq \langle \mathrm{id}_A, \mathrm{id}_A \rangle : A \hookrightarrow A \times A$
- intersection: pullback of $U \hookrightarrow A$ and $V \hookrightarrow A$.

define subobject functor

- pullback well defined
- smallness

greatest subobject as natural transformation $\{*\} \to \mathrm{Sub}$

- pullback of identity
- naturality: essentially, there are only one construction
- in the presence of terminal object, a uniform choice of subobjects $\leftrightarrow$ a subobject of 1
- Yoneda lemma: find the essential one construction, universal element $\mathrm{id}_1 : 1 \hookrightarrow 1$

define subobject classifier

- representable functor: internalisation
- indicator function, unpack naturality
- Yoneda lemma: find the essential one construction
- compute the inclusion map of the universal subobject $T \hookrightarrow \Omega$

translate the greatest subobject

- as natural transformation $\{*\} \to \hom(-, \Omega)$ or as universal element $1 \to \Omega$.
- prove $T \hookrightarrow \Omega$ and $1 \hookrightarrow \Omega$ are the same subobject, back to wiki definition.

outro

- naturally: uniform construction
- representable functor: internalize natural transformation as morphism
- exercise: prove uniform construction of diagonal embedding is NOT natural transformation !!! and prove $x = y$ iff $\mathrm{eq}_A\langle x, y\rangle = \mathrm{true}_\Gamma$, it internalize the equality of morphisom.
- exercise: write the uniform construction of intersection as natural transformation $\mathrm{Sub} \times \mathrm{Sub} \to \mathrm{Sub}$ and as a universal element $\wedge : \Omega \times \Omega \to \Omega$, and prove that $\wedge\,\langle p, q\rangle = \mathrm{true}_\Gamma$ iff $p = \mathrm{true}_\Gamma$ and $q = \mathrm{true}_\Gamma$, it internalize the logic "and".

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intro

subobject classifier seem to be very important to definite the topos, although i also don't know what topos is. in the processes of understanding the definition of subobject classifier, i think i gain more experience in playing around representable functor, yoneda lemma, natural transformation.

this blog will introduce some concept related to the subobject in general category, and see how subobject classifier ***internalize*** these ideas. a lot of category definition relate to representable functor seem to trying internalize some external structure of category, we will see more precise example of it.

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generalize element

let C be a category. in general, for a object A in C, we can't talk about an element of A. but from the category semantic of type theory or in the lens of yoneda lemma. it seem to be a great idea to consider the ***generalize element*** of A.

Definition (generalize element). Let C be a category, A be an object of C. a generalize element of A is a morphism into A

$$x : \Gamma \to A$$

where Γ, the source of x, is called the context of x. in the case C has terminal object 1, and the context of x is 1, x is called a global element of A.

the notation of generalize element make the reason about object very like a "set". the yoneda embedding $\mathrm{y} : \mathcal{C} \to \mathrm{Psh}(\mathcal{C}), A \mapsto \mathrm{hom}_{\mathcal{C}}(-, A)$ can be view as the convert from an object to it's generalize element sorted by contexts (where Psh(C) := Set^(C^op) is the presheaf category of C and we should assume C is locally small so the output of hom(-, A) is a set). the following exercise give more intuition about the generalize element notation:

Exercise: Let C be a category, f, g : X → Y be two morphism in C. show that f = g if and only if fx = gx for all generalize element x of X (where fx denote the composition f ∘ x)

the name of the global element come from the understanding: if we have a global element c : 1 → A, we can form the generalize element $c!_\Gamma : \Gamma \to A$ for each object Γ, where $!_\Gamma$: Γ → 1 denote the unique map into the terminal object. such collection of generalize element $\{c!_\Gamma\}_{\Gamma \in C}$ is invariant under "context substitution" in the sense that $c!_\Gamma \sigma = c!_\Delta$ for all morphism $\sigma : \Delta \to \Gamma$ since $!_\Gamma \sigma = !_\Delta$. conversely, if we have a collection of generalize element $\{c_\Gamma : \Gamma \to A\}_{\Gamma \in C}$ invariant under context substitution i.e. $c_\Gamma \sigma = c_\Delta$ for all morphism $\sigma : \Delta \to \Gamma$, then take $c_1 : 1 \to A$ we have $c_1 !_\Gamma = c_\Gamma$ for each object Γ.

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subobject

For the generalization of a subset, we should look about a monomorphism $i : U \hookrightarrow A$. in the notation of generalized element, the condition of $i : U \hookrightarrow A$ being a monomorphism is: for all generalized element x, y of U with the same context, if ix = iy then x = y (notice ix, iy is a generalized element of A) . so the postcomposition by i can be seen as a inclusion from the generalized element of U to it of A.

we can recover the notation of a element lying a subset, the inclusion relation between two subset and when two subset are consider equal.

Definition: for a monomorphism $u : U \hookrightarrow A$ and a generalized element $x : \Gamma \to A$ of A, we write $x \in u$ if x factor through u, i.e., there exist a morphism $\varphi : \Gamma \to U$ such that x = u φ.

notice such φ is unique since if we have φ, ψ : Γ → U with x = u φ and x = u ψ then u φ = u ψ implies φ = ψ by the monoticity of u. so the notation x ∈ u do not mention about φ is appropriate.

Definition: for monomorphism $u : U \hookrightarrow A, v : V \hookrightarrow A$, we write $u \leq v$ if for all generalized element x of A, $x \in u$ implies $x \in v$. we write u ~ v if u ≤ v and v ≤ u.

the following exercise make should the definition of $u \leq v$ has behaviour we expected and match the definition of in nlab.

Exercise:

1. u ≤ v iff u factor through v.
2. the relation ≤ on the monomorphism into A form a preorder.
3. the relation ~ is a equivalent relation on monomorphism into A.
4. x ∈ u and u ≤ v is well defined with respect to the equivalent relation ~, that is, if u ~ u' then x ∈ u iff x ∈ u'; if u ~ u', v ~ v' then u ≤ v iff u' ≤ v'.

we should closed this section by given the definition of the ***subobjects*** of an object A.

Definition: the subobjects on A is the equivalent class of monomorphisms into A quotient by u ~ v, in notation: $\mathrm{Sub}(A) \coloneqq \{u \mid u : U \hookrightarrow A \text{ monic into } A\}/\sim$. if in a category, the collection Sub(A) are indeed a set for all object A, this category is call well-powered.

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subobject functor and some example

we will look some important examples of subobjects:

Example: Fix an object A, $\mathrm{id}_A : A \hookrightarrow A$ is monic. it represents the maximum subobject of A i.e. for any subobject u of A, we have u ≤ id_A. Or it "collected" all element of A i.e., for any generalized element x of A, we have $x \in \mathrm{id}_A$.

Example: Fix an object A, the diagonal embedding $\Delta_A : A \hookrightarrow A \times A, \Delta_A \coloneqq \langle \mathrm{id}_A, \mathrm{id}_A \rangle$ is monic. it is a important step of internalize the equality of morphism, in the sense that, for all generalize element x, y of A with the same context, we have $x = y$ if and only if $\langle x, y \rangle \in \Delta_A$.

Example: Let f : A → B be a morphism, u : U → B be a subobject of B. a subobject v : V → A of A is called a preimage of u under f, if for all generalize element x of A, x ∈ v iff fx ∈ u.

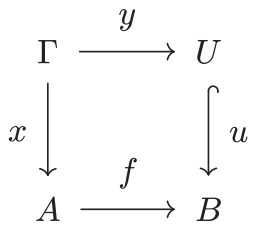
one can check the preimage of u under f is unique (upto subobject equivalent ~) and well-defined (do not depend on the representative of the subobject equivalent class)

- if v, v' are a preimage of u under f, then v ~ v': since x ∈ v iff fx ∈ u iff x ∈ v' for all generalize element x of A.
- if u ~ u', v, v' a the preimage of u, u' under f respectively, then $v \sim v'$: since $x \in v \Leftrightarrow fx \in u \Leftrightarrow fx \in u' \Leftrightarrow x \in v'$ for all generalize element x of A.
  so if a preimage of u under f exist, we can denote it by $f^{-1}(u) : f^{-1}(U) \hookrightarrow A$.

taking preimage is functorial in the sense that: (assume all preimage exist)

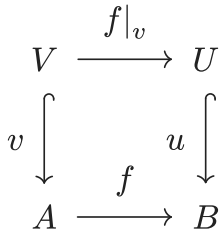
- if u is a subobject of A, then $\mathrm{id}_A^{-1}(u) \sim u$: since $x \in \mathrm{id}_A^{-1}(u) \Leftrightarrow \mathrm{id}_A\, x \in u \Leftrightarrow x \in u$ for all generalized element x of A.
- if u is a subobject of C, f: A → B, g :B → C morphism, then $(gf)^{-1}(u) \sim f^{-1}(g^{-1}(u))$: since $x \in (gf)^{-1}(u) \Leftrightarrow gfx \in u \Leftrightarrow fx \in g^{-1}(u) \Leftrightarrow x \in f^{-1}(g^{-1}(u))$ for all generalized element x of A.

how to show the existence of preimage? for a generalized element $x : \Gamma \to A$, unpack the condition $fx \in u$, we get $fx = uy$ for some $y : \Gamma \to U$, we see a square diagram:



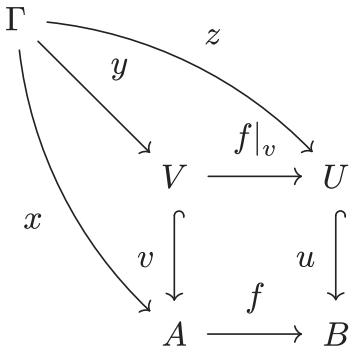
it gives me a strong belief that preimage is related to pullback. In set category, a pullback of a inclusion map of a subset along a function is indeed the preimage. so we try:

if the category C has pullback, Let f : A → B be a morphism, u : U → B be a subobject of B. Consider the pullback square of u along f:



We want to show that, $v : V \hookrightarrow A$ is the preimage of u under f. (notice the pullback of a monomorphism is again a monomorphism, since u is monic, so v is monic too).

Let $x : \Gamma \to A$ is a generalize element of A. See the following diagram for keep track the arrows:



- If x ∈ v, say x = v y for some y : Γ → V, since the pullback diagram commutes, $fx = fvy = uf|_v\ y$ so $fx \in u$.
- Conversely, if f x ∈ u, say f x = u z for some $z : \Gamma \to U$, then (Γ, x, z) from a cone of the diagram $A \overset{f}{\to} B \overset{u}{\hookleftarrow} U$, by the universal property of the pullback V, there exist unique $y : \Gamma \to V$ such that $x = vy$ and $z = f|_v\ y$. so x ∈ v. ∎

Exercise: Conversely, without assume the category has pullback, if $v : V \hookrightarrow A$ is the preimage of u under f, then v = v id_V so v ∈ v, since v is the preimage, f v ∈ u. we get f v = u g for some g : V → U. show that (V, v, g) is the pullback of the diagram $A \overset{f}{\to} B \overset{u}{\hookleftarrow} U$.

We summary this example by the following proposition:

Proposition (Subobject functor): Let C be a well-powered category with pullbacks. We have the subobject functor:

$$\begin{aligned} \mathrm{Sub} : \mathcal{C}^{\mathsf{op}} &\to \mathrm{Set} \\ A &\mapsto \mathrm{Sub}(A) \\ f : A \to B \mapsto f^{-1} &: \mathrm{Sub}(B) \to \mathrm{Sub}(A) \end{aligned}$$

mapping an object A to the set of all subobjects of A. mapping an morphism f: A → B to the preimage function $f^{-1} : \mathrm{Sub}(B) \to \mathrm{Sub}(A)$ which sending a subobject $u \in \mathrm{Sub}(B)$ of B to the unique subobject $f^{-1}(u) \in \mathrm{Sub}(A)$ of A such that $x \in f^{-1}(u) \Leftrightarrow fx \in u$ for all generalized element x of A. Equivalently, $f^{-1}(u)$ is defined as the pullback of u along f:

$$\begin{array}{ccc} f^{-1}(U) & \longrightarrow & U \\ {\scriptstyle f^{-1}(u)}\downarrow & & \downarrow{\scriptstyle u} \\ A & \xrightarrow[f]{} & B \end{array}$$

Key Exercise: If pullback exist. let u be a subobject of A, show that x ∈ u iff $x^{-1}(u) \sim \mathrm{id}_\Gamma$ for all generalized element $x : \Gamma \to A$ of A. it say all proposition like x ∈ u, can be write as some equality of subobject. using the exercise above, x = y iff ⟨x, y⟩ ∈ Δ, we the primary predicte of the language becomes the equality of subobject!

Example: assume the category has pullback. Let u : U → A, v: V → A be a subobject of A. a subobject w of A is a intersection of u and v if x ∈ w iff x ∈ u and x ∈ v. if a intersection exist, and it is unique and well-defined. consider the pullback of v along u:

$$\begin{array}{ccc} u^{-1}(V) & \xrightarrow{u|_v} & V \\ {\scriptstyle u^{-1}(v)}\downarrow & & \downarrow{\scriptstyle v} \\ U & \xrightarrow[u]{} & A \end{array}$$

then the subobject of A defined by the composition $u^{-1}(V) \overset{u^{-1}(v)}{\hookrightarrow} U \overset{u}{\hookrightarrow} A$ is the intersection of u and v, denote by $u \cap v$.

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greatest subobject as natural transformation $\{*\} \to \mathrm{Sub}$

- pullback of identity
- naturality: essentially, there are only one construction
- in the presence of terminal object, a uniform choice of subobjects ↔ a subobject of 1
- Yoneda lemma: find the essential one construction, universal element $\mathrm{id}_1 : 1 \hookrightarrow 1$

now we established the subobject functor, before we step into subobject classifier, i want to change a perceptive to see the greatest subobject $\mathrm{id}_A : A \hookrightarrow A$ again.

the construction of $\mathrm{id}_A : A \hookrightarrow A$ seem do not depend on the object A, we can do this "uniform" construction all over any object. and maybe, there are the only one "essentials" subobject.

this construction are "uniform" in the sense that: we have $\mathrm{id}_A \in \mathrm{Sub}(A)$ for each object A and the functorial of Sub preserve id_A, namely, for any $f : A \to B$, we have $f^{-1}(\mathrm{id}_B) \sim \mathrm{id}_A$.

$$\begin{array}{ccc} B & \mathrm{Sub}(B) & \mathrm{id}_B \\ f \uparrow & f^{-1} \downarrow & \downarrow \\ A & \mathrm{Sub}(A) & \mathrm{id}_A \sim f^{-1}(\mathrm{id}_B) \end{array}$$

indeed, for all generalized element x of A, we have $x = \mathrm{id}_A\, x$ and $fx = \mathrm{id}_B\, fx$, so $x \in \mathrm{id}_A$ and $fx \in \mathrm{id}_B$ i.e. $x \in f^{-1}(\mathrm{id}_B)$ are both hold, so $\mathrm{id}_A \sim f^{-1}(\mathrm{id}_B)$.

this is exactly the ideal of natural transformation! let $\underline{\{*\}} : \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ be the contravariant constant functor sending all object to the singleton set $\{*\}$ and all morphism to identity on singleton set $\mathrm{id}_{\{*\}}$. we can state uniformness of picking one subobject id_A for each object A as: We have a natural transformation defined by:

$$\begin{aligned} \eta &: \underline{\{*\}} \to \mathrm{Sub} \\ \eta_A &: \{*\} \to \mathrm{Sub}(A) \\ \eta_A &: \ * \ \mapsto \mathrm{id}_A \end{aligned}$$

the naturality says for any $f : A \to B$, the following diagram commutes:

$$\begin{array}{ccccc} B & & \{*\} & \xrightarrow{\eta_B} & \mathrm{Sub}(B) \\ f \uparrow & & \mathrm{id}_{\{*\}} \| & & f^{-1} \downarrow \\ A & & \{*\} & \xrightarrow{\eta_A} & \mathrm{Sub}(A) \end{array} \qquad \begin{array}{ccc} * & \longmapsto & \mathrm{id}_B \\ \downarrow & & \downarrow \\ * & \longmapsto & \mathrm{id}_A \sim f^{-1}(\mathrm{id}_B) \end{array}$$

so $\mathrm{id}_A \sim f^{-1}(\mathrm{id}_B)$!

the naturality given described the "uniform" part, and the Yoneda lemma given help us to find the "essential" core. recall the definition of terminal object 1, it has the property that, for any object A, we have one and only one arrow $!_A : A \to 1$ from A into 1, in other word, $\hom_{\mathcal{C}}(A, 1)$ is a singleton set, so we have a set bijection

$$\{*\} \cong \hom_{\mathcal{C}}(A, 1), * \mapsto !_A$$

more over, this bijection is natural in A, (the assignment of $!_A$ for each object A is also uniform), that is, the following defined a natural transformation (in fact, natural isomorphism, since every component is bijection)

$$\xi : \underline{\{*\}} \cong \hom_{\mathcal{C}}(-, 1)$$
$$\xi_A : \{*\} \cong \hom_{\mathcal{C}}(A, 1)$$
$$\xi_A : \ * \ \mapsto !_A$$

one can check, using the fact the functorial of $\hom_{\mathcal{C}}(-,1)$ is the postcomposition by $(\circ f) : \hom_{\mathcal{C}}(B,1) \to \hom_{\mathcal{C}}(A,1)$, the naturality of ξ is $!_A = !_B f$ for all $f : A \to B$. composite them, we get a natural transformation:

$$\theta := \eta\xi^{-1} : \hom_{\mathcal{C}}(-,1) \overset{\xi^{-1}}{\cong} \underline{\{*\}} \overset{\eta}{\to} \mathrm{Sub}$$

whose component at A is $\hom_{\mathcal{C}}(A,1) \to \mathrm{Sub}(A), !_A \mapsto \mathrm{id}_A$.

think about how to make a uniform mapping that send a morphism $f : A \to 1$ to a subobject $u_f \in \mathrm{Sub}(A)$ of A, and this mapping need to work for all object A. if we have a subobject $v \in \mathrm{Sub}(1)$ of 1, we can use the functorial of Sub to get $f^{-1}(v) \in \mathrm{Sub}(A)$. it turns out is the only way to make a such uniform mapping! the trick is, we can use $\mathrm{id}_1 : 1 \to 1$ to detect such uniform mapping!

consider the component of $\theta : \hom_{\mathcal{C}}(-,1) \to \mathrm{Sub}$ at 1 and input id_1:

$$\theta_1 : \hom_{\mathcal{C}}(1,1) \to \mathrm{Sub}(1)$$
$$\theta_1 : \mathrm{id}_1 = !_1 \mapsto \mathrm{id}_1$$

(the left hand side $\mathrm{id}_1 : 1 \to 1$, the right hand side $\mathrm{id}_1 : 1 \hookrightarrow 1$ is a subobject of $\mathrm{Sub}(1)$)
the naturality recover all component from the "essential" subobject $\mathrm{id}_1 \in \mathrm{Sub}(1)$: for any object A, we have the morphism $!_A : A \to 1$, by naturality, the following diagram commutes:

$$\begin{array}{ccc} 1 & \hom_{\mathcal{C}}(1,1) \xrightarrow{\theta_1} \mathrm{Sub}(1) & \mathrm{id}_1 = !_1 \longmapsto \mathrm{id}_1 \\ {\scriptstyle !_A}\uparrow & \downarrow{\scriptstyle (\circ\, !_A)} \quad\quad {\scriptstyle !_A^{-1}}\downarrow & \downarrow \quad\quad\quad\quad \downarrow \\ A & \hom_{\mathcal{C}}(A,1) \xrightarrow{\theta_A} \mathrm{Sub}(A) & \mathrm{id}_1\, !_A = !_A \longmapsto \mathrm{id}_A \sim !_A^{-1}(\mathrm{id}_1) \end{array}$$

so $\mathrm{id}_A \sim !_A^{-1}(\mathrm{id}_1) \in \mathrm{Sub}(A)$. all greatest subobject $\mathrm{id}_A \in \mathrm{Sub}(A)$ are the *shadow* of the "essential" core $\mathrm{id}_1 \in \mathrm{Sub}(1)$ in the way id_A are preimage of id_1 under $!_A$ for all object A.

this process of find the "essential" core are basically the proof of yoneda lemma and the representable functor. we state three theorem and definition that we will use in the following paragraph. i suggested also see nlab to find more detail. but also don't worry about these statement, we will unpack the proof in the most use of these theorem.

Theorem (Yoneda Lemma): Let $\mathcal{C}$ be a locally small category. Let $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ be a contravariant set functor, X be an object in $\mathcal{C}$. Then we have a bijection

$$\hom_{\mathrm{Psh}(\mathcal{C})}(\hom_{\mathcal{C}}(-, X), F) \cong F(X)$$

which send a natural transformation $\eta \in \hom_{\mathrm{Psh}(\mathcal{C})}(\hom_{\mathcal{C}}(-, X), F)$ i.e. $\eta : \hom_{\mathcal{C}}(-, X) \to F$ to the element $\eta_X(\mathrm{id}_X) \in F(X)$ (the essential core!). where $\mathrm{Psh}(\mathcal{C}) \coloneqq \mathrm{Set}^{\mathcal{C}^{\mathsf{op}}}$ denote the presheaf category which object are functor, morphism are natural transformation. we also have a version for covariant $G : \mathcal{C} \to \mathrm{Set}$ set functor.

Definition (Representable functor): Let $\mathcal{C}$ be a locally small category. A functor $F : \mathcal{C}^{\mathsf{op}} \to \mathrm{Set}$ is called representable if there exist some object X in $\mathcal{C}$ and a natural isomorphism

$$\theta : \hom_{\mathcal{C}}(-, X) \cong F$$

in this case, X called the representing object of F. By Yoneda lemma, we have an element $e \in F(X)$ unique determine θ, is called the universal element of F. we also have a version for covariant $G : \mathcal{C} \to \mathrm{Set}$ set functor.

Corollary (Yoneda embedding): Let $\mathcal{C}$ be a locally small category and X, Y be two object of $\mathcal{C}$. then if we have a isomorphism

$$\hom_{\mathcal{C}}(\Gamma, X) \cong \hom_{\mathcal{C}}(\Gamma, Y)$$

natural in Γ (i.e. a natural isomorphism $\hom_{\mathcal{C}}(-, X) \cong \hom_{\mathcal{C}}(-, Y)$), then $X \cong Y$. in the notation of generalized element, informally, it says, if all the generalized element of X and Y are the same, the X and Y are the same.

Exercise: In starting paragraph, we explained why a morphism from terminal object $1 \to A$ called a global object. Revised that section, and rewrite the condition "invariant under context substitution" as natural transformation and show that we have the bijection

$$\hom_{\mathrm{Psh}(\mathcal{C})}\Big(\underline{\{*\}}, \hom_{\mathcal{C}}(-, A)\Big) \cong \hom_{\mathcal{C}}(1, A)$$

(bonus, it is also natural in A)

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define subobject classifier

- representable functor: internalisation
- indicator function, unpack naturality
- Yoneda lemma: find the essential one construction
- compute the inclusion map of the universal subobject $T \hookrightarrow \Omega$

now we have enough courage to see the definition of subobject classifier. from now on, we will assume our category $\mathcal{C}$ has pullback, terminal object, is locally small and well-powered.

Definition: The subobject classifier Ω is the representing object of subobject functor Sub. Explicitly, Ω is a object in $\mathcal{C}$ and we have a natural isomorphism:

$$\chi : \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega)$$

from above discussion, we have expected the naturality of χ gives we a uniform way to convert a subobject of A to a morphism into Ω. we may also use yoneda lemma to find the essential core of χ. more over, the isomorphism part of χ say all subobject are totally characterised by the arrow into Ω. we encourage the reader to play around the naturality of χ and see how the example in above paragraph corresponded to, you may guess all the construction we will do in the below section.

Let's unpack the definition of subobject classifier

$$\begin{aligned}\chi : \quad & \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega) \\ \chi_A : \mathrm{Sub}(A) & \cong \hom_{\mathcal{C}}(A, \Omega) \\ \chi_A : \quad & u \mapsto \chi_A(u) =: \chi_u\end{aligned}$$

first, χ is a natural transformation it has components $\chi_A : \mathrm{Sub}(A) \to \hom_{\mathcal{C}}(A, \Omega)$ for each object A. let's denote the image of a subobject $u \in \mathrm{Sub}(A)$ under χ_A as $\chi_u := \chi_A(u)$, notice we can infer the object A by the target of the monomorphism $u : U \hookrightarrow A$, so the notation χ_u do not mention about A is appropriate. we call χ_u the indicator (or characteristic) function of u.

the naturality of χ gives (can you guess what it is by just recall the functorial of Sub and $\hom_{\mathcal{C}}(-, \Omega)$ without drawing the natural square?), for any morphism $f : A \to B$, we have

$$\begin{array}{ccc} B & \mathrm{Sub}(B) & \xrightarrow{\chi_B} \hom_{\mathcal{C}}(B, \Omega) \\ f \uparrow & f^{-1} \downarrow & (\circ f) \downarrow \\ A & \mathrm{Sub}(A) & \xrightarrow{\chi_A} \hom_{\mathcal{C}}(A, \Omega) \end{array} \qquad \begin{array}{ccc} u & \longmapsto & \chi_u \\ \downarrow & & \downarrow \\ f^{-1}(u) & \longmapsto & \chi_{f^{-1}(u)} = \chi_u f \end{array}$$

commutes, that is, $\chi_{f^{-1}(u)} = \chi_u f$ for any subobject $u \in \mathrm{Sub}(B)$. (compare it to set-theoretic indicator function of a subset!)

Since χ is isomorphism, all morphism into Ω should be the indicator function for some subobject, in fact, we can compute this subobject explicitly: Yoneda lemma tell us, we should consider the universal subobject $t : T \hookrightarrow \Omega \in \mathrm{Sub}(\Omega)$ defined by

$$\begin{aligned}\chi_\Omega : \mathrm{Sub}(\Omega) & \cong \hom_{\mathcal{C}}(\Omega, \Omega) \\ \chi_\Omega : \quad & t \mapsto \mathrm{id}_\Omega\end{aligned}$$

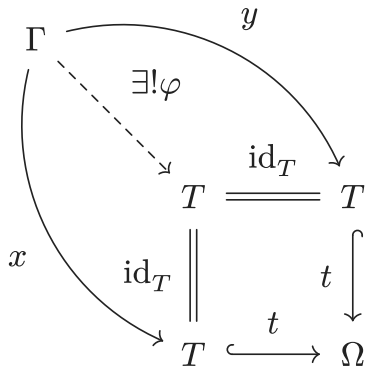
For any morphism $f : A \to \Omega$, apply naturality to $f : A \to \Omega$ and $t \in \mathrm{Sub}(\Omega)$ then

$$\begin{aligned}\chi_{f^{-1}(t)} &= \chi_t f \\ &= \mathrm{id}_\Omega \, f \\ &= f\end{aligned}$$

that is, $\hom_{\mathcal{C}}(A, \Omega) \to \mathrm{Sub}(A), f \mapsto f^{-1}(t)$ give us the inverse of χ. all subobjects just a preimage of the universal subobject t!

Key Exercise: Show that $x \in u \Leftrightarrow \chi_u x \in t$ for any subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A. it gives the justification of the name of χ_u is call indicator function and t is the first alphabet of "true".

Finally, we can also compute the monomorphism $t : T \hookrightarrow \Omega$ as an indicator function of some subobject. for the sake of showing the proof technical of category theory, we compute $t^{-1}(t)$ in three ways:

- compute $t^{-1}(t)$ by pullback: consider a cone $T \overset{x}{\leftarrow} \Gamma \overset{y}{\rightarrow} T$ of the diagram $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$:
$$\begin{array}{ccc} \Gamma & \xrightarrow{y} & T \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle t} \\ T & \xrightarrow{t} & \Omega \end{array}$$
we have two data x, y and one constraint, $tx = ty$. since t is monic, $x = y$. so a cone of $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$ really has one data $x : \Gamma \to T$ without any constraint; it holds the same data as a cone of the diagram consisting of a single object T. but the limit of the diagram that consists of a single object T is T with the universal cone $\mathrm{id}_T : T \to T$. so we claim: the following is a pullback diagram
$$\begin{array}{ccc} T & \xrightarrow{\mathrm{id}_T} & T \\ {\scriptstyle \mathrm{id}_T}\downarrow & & \downarrow{\scriptstyle t} \\ T & \xrightarrow{t} & \Omega \end{array}$$
indeed, it commutes $t\,\mathrm{id}_T = t\,\mathrm{id}_T$. given any cone $T \overset{x}{\leftarrow} \Gamma \overset{y}{\rightarrow} T$ with $tx = ty$:


since $tx = ty$ and t is monic, $x = y$.
- existence of φ: take $\varphi \coloneqq x : \Gamma \to T$ and we have $x = \mathrm{id}_T\, x$ and $y = x = \mathrm{id}_T\, x$.
- uniqueness of φ: if $\psi : \Gamma \to T$ such that $x = \mathrm{id}_T\, \psi$ and $y = \mathrm{id}_T\, \psi$ than $x = \mathrm{id}_T\, \psi = \psi$. (in general, a pullback of monomorphism of itself is the identity map.)

Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

- compute $t^{-1}(t)$ by Yoneda lemma: in general, the existence of pullback of the diagram $A \overset{f}{\rightarrow} C \overset{g}{\leftarrow} B$ is exactly to say we have an object $A \times_C B$ and a isomorphism

$$\hom_{\mathcal{C}}(\Gamma, A \times_C B) \cong \{(x, y) \in \hom_{\mathcal{C}}(\Gamma, A) \times \hom_{\mathcal{C}}(\Gamma, B) \mid fx = gy\}$$

natural in Γ. don't be scary about this statement. think about the functorial of functors at both side (verify the right hand side is a functor in Γ! it send a object Γ to the collection of cones of the diagram $A \xrightarrow{f} C \xleftarrow{g} B$ out from Γ, and functorial is act by precomposition). follow the process above to unpack the naturality and find the universal element, you will see it is exact the universal property of pullback.)

To compute the pullback $T \times_\Omega T$ of the diagram $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$. the machinery of Yoneda lemma and representable functor, enable us to do the following computation:

$$\begin{aligned}
&\hom_{\mathcal{C}}(\Gamma, T \times_\Omega T) \\
\cong &\{(x, y) \in \hom_{\mathcal{C}}(\Gamma, T) \times \hom_{\mathcal{C}}(\Gamma, T) \mid tx = ty\} \\
\cong &\{(x, y) \in \hom_{\mathcal{C}}(\Gamma, T) \times \hom_{\mathcal{C}}(\Gamma, T) \mid x = y\} \\
\cong &\{x \in \hom_{\mathcal{C}}(\Gamma, T) \mid \mathrm{True}\} \\
= &\hom_{\mathcal{C}}(\Gamma, T)
\end{aligned}$$

is an isomorphism natural in Γ (why?). So by Yoneda lemma, $T \times_\Omega T \cong T$ and the univeral cone is given by the element corresponding to the $\mathrm{id}_T \in \hom_{\mathcal{C}}(T, T)$ (why?):

$$\begin{aligned}
&\{(x, y) \in \hom_{\mathcal{C}}(T, T) \times \hom_{\mathcal{C}}(T, T) \mid tx = ty\} &\ni (\mathrm{id}_T, \mathrm{id}_T) \\
\cong &\{(x, y) \in \hom_{\mathcal{C}}(T, T) \times \hom_{\mathcal{C}}(T, T) \mid x = y\} &\ni (\mathrm{id}_T, \mathrm{id}_T) \\
\cong &\{x \in \hom_{\mathcal{C}}(T, T) \mid \mathrm{True}\} &\ni \mathrm{id}_T \\
= &\hom_{\mathcal{C}}(T, T) &\ni \mathrm{id}_T
\end{aligned}$$

so the following is a pullback square

$$\begin{array}{ccc}
T & \xlongequal{\mathrm{id}_T} & T \\
{\scriptstyle \mathrm{id}_T}\,\Big\| & & \Big\downarrow{\scriptstyle t} \\
T & \xhookrightarrow{t} & \Omega
\end{array}$$

Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

- compute $t^{-1}(t)$ by generalized element: by definition of preimage, for all generalized element x of T, we have
$$x \in t^{-1}(t) \Leftrightarrow tx \in t$$
but $tx = tx$ so $tx \in t$ for any x. so $t^{-1}(t)$ is the greatest subobject in $\mathrm{Sub}(t)$. Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

the comparison of this three proof is very interesting. i personal like the second method the most, Yoneda lemma gives we a strong, flexible and intuitive way to think about an object defined as some representing object of a functor. it includes example like limit, exponential object, subobject classifier and more. but in principle, it may be wordy: one need to justify every

step in the computation is natural and isomorphism, and write down how elements were send and check properties.

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translate the greatest subobject

- as natural transformation $\{*\} \to \hom(-, \Omega)$ or as universal element $1 \to \Omega$.
- prove $T \hookrightarrow \Omega$ and $1 \hookrightarrow \Omega$ are the same subobject, back to wiki definition.

In previous section, we see the subobject classifier gives the following data

- an universal subobject $t : T \hookrightarrow \Omega$. (in fact, $t = \chi_{\mathrm{id}_T} : T \to \Omega$)
- for any subobject $u \in \mathrm{Sub}(A)$, we have an unique morphism $\chi_u : A \to \Omega$ such that $u \sim \chi_u^{-1}(t)$.

in this section, we want to see how subobject classifier ***internalize*** some special subobjects into morphisms in the underline category. Externally, we have concepts:

- the equality of morphism $f = g$,
- the membership of an generalized element $x \in u$ are in a subobject,
- the logic "and", or the intersection subobject $u \cap v$.
  we claim that, we can convert each these concept as an morphism $p : A \to \Omega$ and we have an collection of specified morphsim called $\mathrm{true}_A : A \to \Omega$ for each A, such that $p = \mathrm{true}_A$ are equivalent to $f = g$, $x \in u$, or $q_1 = \mathrm{true}_A \wedge q_2 = \mathrm{true}_A$.

first, we focus on finding what is the true.

recall the greatest subobject $\mathrm{id}_A : A \hookrightarrow A \in \mathrm{Sub}(A)$, we seen this construction are so uniform such that the assignment

$$\begin{aligned} \underline{\{*\}} &\to \mathrm{Sub} \\ \{*\} &\to \mathrm{Sub}(A) \\ * &\mapsto \mathrm{id}_A \end{aligned}$$

form a natural transformation. now, using the concept of representable functor, the constant singleton set $\underline{\{*\}}$ is represented by the terminal object 1, and the subobject functor Sub is represented by the subobject classifier Ω, so we have

$$\hom_{\mathcal{C}}(-, 1) \cong \underline{\{*\}} \to \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega)$$

it send $!_A \mapsto * \mapsto \mathrm{id}_A \mapsto \chi_{\mathrm{id}_A}$ and naturality say $\chi_{\mathrm{id}_B} = \chi_{\mathrm{id}_A} f$ for all morphism $f : B \to A$.
by Yoneda lemma, compute $\mathrm{id}_1 \mapsto * \mapsto \mathrm{id}_1 \mapsto \chi_{\mathrm{id}_1}$, we get an universal element $\chi_{\mathrm{id}_1} : 1 \to \Omega$ such that $\chi_{\mathrm{id}_A} = \chi_{\mathrm{id}_1} !_A$ for all object A.

Key Exercise: Notice that for a morphism pointing out from terminal object is a monomorphism, so $\chi_{\mathrm{id}_1} : 1 \hookrightarrow \Omega \in \mathrm{Sub}(\Omega)$. Show that $x \in u \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$ for any subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A.

if you did't shoot out WOW here, you should go back and do all of three Key Exercises above.

spoiler warning!!!

both $t, \chi_{\mathrm{id}_1} \in \mathrm{Sub}(\Omega)$ has the property that, for all $u \in \mathrm{Sub}(A)$ generalized element $x : \Gamma \to A$ of A, we have

$$x \in u \Leftrightarrow \chi_u x \in t$$
$$x \in u \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$$

both act like the "true"! indeed, take $u \coloneqq t$ in second equation, we have $x \in t \Leftrightarrow \chi_t x \in \chi_{\mathrm{id}_1}$ but $\chi_t = \mathrm{id}_\Omega$ by construction, so $t \sim \chi_{\mathrm{id}_1} \in \mathrm{Sub}(\Omega)$ they are indeed the same subobject!

proof of equation 1: by the property of t, we have $u \sim \chi_u^{-1}(t)$. expand the definition of $\sim$ and preimage, we are done.
proof of equation 2: first, we have $x \in u \Leftrightarrow x^{-1}(u) \sim \mathrm{id}_\Gamma$, since

$$x \in u \Leftrightarrow x\,\mathrm{id}_\Gamma \in u \Leftrightarrow \mathrm{id}_\Gamma \in x^{-1}(u) \Leftrightarrow \mathrm{id}_\Gamma \leq x^{-1}(u) \Leftrightarrow \mathrm{id}_\Gamma \sim x^{-1}(u)$$

where we used $\mathrm{id}_\Gamma \leq x^{-1}(u)$ iff id_Γ factor through $x^{-1}(u)$ and id_Γ is the greatest subobject.
next, we will show $x^{-1}(u) \sim \mathrm{id}_\Gamma \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$. if $x^{-1}(u) \sim \mathrm{id}_\Gamma$, take χ at both side, we get

$$\begin{array}{ccc} \chi_{x^{-1}(u)} & = & \chi_{\mathrm{id}_\Gamma} \\ \| & & \| \\ \chi_u x & & \chi_{\mathrm{id}_1} !_\Gamma \end{array}$$

where two vertical equality use the naturality of $u \mapsto \chi_u$ and $!_A \mapsto \chi_{\mathrm{id}_A}$. therefore, $\chi_u x \in \chi_{\mathrm{id}_1}$. Conversely, if $\chi_u x \in \chi_{\mathrm{id}_1}$, say $\chi_u x = \chi_{\mathrm{id}_1} y$ for some $y : \Gamma \to 1$. again, by naturality, $\chi_{x^{-1}(u)} = \chi_u x = \chi_{\mathrm{id}_1} y = \chi_{\mathrm{id}_\Gamma}$, therefore $x^{-1}(u) \sim \mathrm{id}_\Gamma$.

We should give the universal subobject a new name $\mathrm{true} : 1 \hookrightarrow \Omega, \mathrm{ture} \coloneqq \chi_{\mathrm{id}_1}$ then $\mathrm{true} \sim t \in \mathrm{Sub}(\Omega)$. it can be understand as a global element of Ω. so it also correspond to a collection of morphism $\mathrm{true}_A : A \to \Omega$ (in fact, $\mathrm{true}_A = \chi_{\mathrm{id}_A}$, $\mathrm{true}_1 = \mathrm{true}$) which is invariant under context substitution: $\mathrm{true}_\Gamma = \mathrm{true}_\Delta \sigma$ for all $\sigma : \Delta \to \Gamma$ and so

$$\mathrm{true}_A = \mathrm{true}\ !_A$$

for all object A.

Finally, rewrite above results, we have

$$x \in u \Leftrightarrow \chi_u x = \mathrm{true}\ !_\Gamma$$

for all subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A. Equivalently, the following

$$\begin{array}{ccc} U & \xrightarrow{!_U} & 1 \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \text{true}} \\ A & \xrightarrow{\chi_u} & \Omega \end{array}$$

is a pullback square, for all subobject $u \in \mathrm{Sub}(A)$.

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outro

- naturally: uniform construction
- representable functor: internalize natural transformation as morphism
- exercise: prove uniform construction of diagonal embedding is NOT natural transformation !!! and prove $x = y$ iff $\mathrm{eq}_A\langle x, y\rangle = \mathrm{true}_\Gamma$, it internalize the equality of morphisom.
- exercise: write the uniform construction of intersection as natural transformation $\mathrm{Sub} \times \mathrm{Sub} \to \mathrm{Sub}$ and as a universal element $\wedge : \Omega \times \Omega \to \Omega$, and prove that $\wedge\,\langle p, q\rangle = \mathrm{true}_\Gamma$ iff $p = \mathrm{true}_\Gamma$ and $q = \mathrm{true}_\Gamma$, it internalize the logic "and".

...

Exercise: Recall that we have the diagonal embedding subobject $\Delta_A : A \hookrightarrow A \times A, \Delta_A \coloneqq \langle \mathrm{id}_A, \mathrm{id}_A\rangle$ satisfies $x = y$ if and only if $\langle x, y\rangle \in \Delta_A$ for all generalize element $x, y : \Gamma \to A$. defined $\mathrm{eq}_A : A \times A \to \Omega, \mathrm{eq}_A \coloneqq \chi_{\Delta_A}$. show that

$$x = y \Leftrightarrow \mathrm{eq}_A\langle x, y\rangle = \mathrm{true}_\Gamma$$

we can understand $\mathrm{eq}_A\langle x, y\rangle : \Gamma \to \Omega$ as the internal proposition means x, y are equal or not. Show that

$$\begin{aligned} \underline{\{*\}} &\to \mathrm{Sub}(- \times -) \\ \{*\} &\to \mathrm{Sub}(A \times A) \\ * &\mapsto \Delta_A \end{aligned}$$

is NOT an natural transformation (why!?).

Exercise: Recall that we have the intersection $u \cap v \in \mathrm{Sub}(A)$ for any subobjects $u, v \in \mathrm{Sub}(A)$, satisfying $x \in u \cap v \Leftrightarrow x \in u \wedge x \in v$. show that

$$\begin{aligned} \mathrm{Sub} \times \mathrm{Sub} &\to \mathrm{Sub} \\ \mathrm{Sub}(A) \times \mathrm{Sub}(A) &\to \mathrm{Sub}(A) \\ (u, v) &\mapsto u \cap v \end{aligned}$$

is a natural transformation. use $\mathrm{Sub} \cong \mathrm{hom}_{\mathcal{C}}(-, \Omega)$ and Yoneda lemma, show it induce a morphism $\mathrm{and} : \Omega \times \Omega \to \Omega$ satisfy:

$$\mathrm{and}\langle p, q\rangle = \mathrm{true}_\Gamma \Leftrightarrow (p = \mathrm{true}_\Gamma \wedge q = \mathrm{true}_\Gamma)$$

for all morphism $p, q : \Gamma \to \Omega$. we should call a morphism $p : \Gamma \to \Omega$ a internal predicate on Γ. we can understand $\mathrm{and} : \Omega \times \Omega \to \Omega$ as the internal connectives "and".